MODULE 5A · EPISODE 2

The Earth’s Atmosphere: Weather and Climate Systems

The Physics Behind the Weather That Surrounds Us

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

The Earth’s Atmosphere: Weather and Climate Systems

This episode descends from the layers explored in Episode 1 into the troposphere, where all of Earth’s weather happens. The panel treats weather as physics: thermodynamic events driven by heat moving from warmer places to cooler ones. It works through six ingredients in turn: the atmosphere itself; water, with its precipitation, hail, lightning, clouds, evaporation, and humidity; the Sun, with albedo and greenhouse gases; Earth’s rotation, with convection cells, the Coriolis Effect, and global winds; volcanic activity; and biological activity, closing with how human activity is changing the climate.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist, atmospheric physics

Learning Objectives

After this episode students should be able to:

- Describe weather as thermodynamic events in the troposphere and name its six main ingredients.
- Explain how rain, snow, sleet, and hail form, and how charge separation in a cloud produces lightning and thunder.
- Use the flash-to-thunder delay to estimate the distance to a lightning strike.
- Explain how clouds form around condensation nuclei, why evaporation cools, and how latent heat feeds a cloud.
- Define relative humidity and dew point, and use them to explain dew and dry winter indoor air.
- Explain albedo and the greenhouse effect, and why nitrogen and oxygen are not greenhouse gases.
- Connect differential heating, convection cells, and the Coriolis Effect to global wind patterns.
- Define temperature and convert between the Fahrenheit and Celsius scales.
- Compare the short-term and long-term climate effects of volcanoes with the effects of human activity.

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QUICK REFERENCE

Key Terms

Term	Definition
Weather	Thermodynamic events in the troposphere, driven by the movement of heat due to temperature differences.
Thermodynamic event	An event driven by the movement of heat from warmer to cooler regions.
Meteorology	The study of weather; from the Greek *meteoros*, "lofty" or "high in the air."
Precipitation	Water falling from the sky as rain, snow, sleet, or hail.
Sleet	Small pellets of ice that form when falling raindrops freeze before reaching the ground.
Hail	Layered ice balls built up as updrafts carry ice particles through supercooled water droplets.
Graupel	Soft hail pellets; collisions between graupel and ice crystals separate electric charge in a cloud.
Lightning	A giant electrical discharge that releases the voltage difference built up in a storm cloud.
Thunder	The sound of the shock wave made when lightning heats the air so suddenly that it expands explosively.
Condensation nuclei	Tiny airborne particles (dust, smoke, sulfates, bacteria) around which cloud droplets and ice crystals form.
Evaporation	The persistent transfer of fast-moving water molecules from liquid water into the air above it; a cooling process.
Latent heat	Energy carried off by evaporating water and released again when the vapor condenses.
Relative humidity	Current water vapor pressure divided by the saturation vapor pressure, times 100%.
Saturation vapor pressure	The maximum water vapor pressure air can hold at a given temperature; rises with temperature.
Dew point	The temperature to which air must cool for its relative humidity to reach 100%.
Homosphere	The well-mixed lower atmosphere: troposphere, stratosphere, and mesosphere.
Albedo	Earth's reflectivity; the fraction of incoming sunlight reflected straight back to space (about 30%).
Greenhouse gas	A gas that absorbs and emits infrared radiation, such as carbon dioxide, methane, water vapor, nitrous oxide, and ozone.
Differential heating	Uneven heating of Earth because sunlight strikes the equator more directly than the poles.

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Term	Definition
Convection cell	A circular loop of air formed as warm air rises and cool air sinks.
Hadley, Ferrel, and polar cells	The three convection cells in each hemisphere, from equator to pole.
Coriolis Effect	An apparent force from Earth's rotation; deflects moving air to its right in the Northern Hemisphere.
Trade winds	Surface winds of the Hadley cell, deflected toward the west by the Coriolis Effect.
Temperature	An indirect measure of the average kinetic energy of a substance's randomly moving molecules.
Sulfate aerosols	A haze of tiny sulfate droplets formed from volcanic sulfur dioxide; reflects sunlight for a year or two.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
6	Main ingredients of Earth-based weather
5 mm to 15+ cm	Range of hailstone sizes (pea-sized to larger than a softball)
8 in (about 20 cm)	Largest hailstone on record in the United States
1 second	Of flash-to-thunder delay is about 1/5 mile, or about 1/3 kilometer
100%	Relative humidity at which cloud droplets form (air at its dew point)
10	Basic cloud types (genera) recognized by meteorologists
About 30%	Share of incoming sunlight Earth reflects straight back into space
3 or more	Atoms per molecule in nearly all greenhouse gases
About 60 °F	How much colder Earth's average surface would be without greenhouse gases
About 2.4 billion years ago	Earth likely froze over almost completely
3 per hemisphere	Convection cells: Hadley, Ferrel, and polar
32 °F and 212 °F	Freezing and boiling points of water at sea level, Fahrenheit
0 °C and 100 °C	Freezing and boiling points of water at sea level, Celsius
C = 5/9 (F − 32)	Fahrenheit-to-Celsius conversion
1991	Eruption of Mount Pinatubo, Philippines; cooled Earth about 0.5 C° (about 1 F°) the next year
1 to 2 years	How long volcanic sulfate aerosols linger in the stratosphere

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Value	What it refers to
About 1%	Volcanic CO_2 (a few hundred million metric tons per year) compared with human CO_2
150 years	Span of temperature records showing human-driven warming
1 to 3 ft (0.3 to 1 m)	Projected sea-level rise this century

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — What Weather Is

All Earth-based weather happens in the **troposphere**, so when we say "atmosphere" in the context of weather, we mean that lowest layer. From a physics standpoint, weather is a set of **thermodynamic events**: events driven by the movement of heat due to temperature differences. Weather is essentially the result of **energy transfer and energy transformation**.

Six main ingredients combine to create Earth-based weather: the atmosphere itself, water in all three phases, the Sun, Earth's rotation, volcanic activity, and biological activity.

Meteorology, the study of weather, comes from the Greek *meteoros*, "lofty" or "high in the air," so it literally means "the study of things high in the air." Without an atmosphere there is no weather. **The Moon** receives sunlight and swings through large temperature changes, but it lacks a substantial atmosphere to create the fluid dynamics weather needs.

Part 2 — Water: Precipitation, Hail, Lightning, and Thunder

Water is the **only common substance that naturally exists in all three phases** (ice, liquid water, and water vapor) within the temperatures and pressures of Earth's atmosphere, which is why it plays perhaps the most crucial role in weather.

Precipitation falls as liquid rain, as snow (fully formed ice crystal structures), or as **sleet**, small pellets of ice that form when falling raindrops freeze before they reach the ground. In violent storms with **cumulonimbus clouds**, strong updrafts carry ice particles up and down through supercooled water droplets that freeze on layer by layer, building **hailstones** from about **5 millimeters to more than 15 centimeters** across. The largest on record in the United States measured **8 inches**, about 20 centimeters.

Inside a thunderstorm cloud, collisions between ice crystals and **graupel**, soft hail pellets, create electric charge separations and enormous voltage differences, within the cloud and between the cloud and the ground. **Lightning** is the giant electrical discharge that dissipates that voltage difference, heating and ionizing the air along its path. It can strike within a cloud, cloud to cloud, cloud to ground, or even ground to cloud. Its energy breaks up nitrogen molecules, forming compounds that dissolve in rainwater as **nitrates** and help fertilize vegetation below.

Thunder is the sound of the shock wave made when lightning heats the air so suddenly that it expands explosively. Because sound travels much more slowly than light, each second between the flash and the thunder means about **one-fifth of a mile**, or about a third of a kilometer.

Part 3 — Clouds and Evaporation

Clouds form from very tiny water droplets or ice crystals that gather around **condensation nuclei**: tiny airborne specks of dust, smoke, sulfate particles, and bacteria. They form where the **relative humidity reaches 100%**, because the air has cooled to the **dew point** for air at that altitude.

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Wind, cold air fronts, or mountains lift and cool moist air. The droplets grow by colliding and merging, or by feeding the growth of ice crystals, until they are too heavy to stay aloft and precipitation begins. Clouds can appear throughout the **homosphere** (troposphere, stratosphere, and mesosphere), but almost all form in the troposphere. Meteorologists classify clouds into **ten basic types, called genera**, and **fog** is simply a cloud at ground level.

Water gets into the air by **evaporation**, the persistent transfer of fast-moving water molecules from liquid water to the air above it. The fastest molecules escape and take their kinetic energy with them, so **evaporation is a cooling process**. The rate depends on the puddle's size and depth, the warmth of the air and ground, wind, direct sunlight, and how low the surrounding relative humidity is.

When the vapor condenses into cloud droplets and ice crystals, it releases that energy as **latent heat**, warming the air. The warmer, more buoyant air fuels **updrafts** that keep the cloud aloft and growing.

† The episode says latent heat warms the air "directly below" the cloud's droplets. More precisely, the heat is released where condensation happens, warming the air inside the cloud and the rising air that feeds it. That extra buoyancy is what drives the updrafts.

Part 4 — Humidity and Dew Point

Relative humidity is the ratio of the current water vapor pressure to the saturation water vapor pressure, as a percentage: **RH = (current vapor pressure ÷ saturation vapor pressure) × 100%**. The **saturation vapor pressure** depends almost entirely on temperature; the warmer the air, the higher it is, so warmer air has more "elbow room" for water vapor.

If air below 100% relative humidity cools enough, its relative humidity reaches 100%; that temperature is the **dew point**. If the same air warms instead, its relative humidity falls.

That explains everyday effects. **Dew** forms overnight when grass, bushes, or a car cool to the dew point of the air around them. In **winter**, cold outside air holds very little water vapor even near 100% relative humidity, so when it is warmed indoors its relative humidity drops sharply. A **drought** is prolonged below-average precipitation in one location.

Part 5 — The Sun, Albedo, and Greenhouse Gases

Heating that drives weather comes almost exclusively from **the Sun**, a nearly constant, renewable source of radiant energy. Some of it warms the atmosphere directly; the rest warms lands and seas, which in turn warm the atmosphere.

About **30%** of incoming sunlight is reflected straight back into space, a fraction set by Earth's **albedo**, its reflectivity. How much of the rest Earth holds onto depends on **greenhouse gases**, gases that absorb and emit infrared radiation and send part of it back toward the surface. Nearly all have **three or more atoms per molecule**, so the two-atom nitrogen and oxygen that make up most of our air are not greenhouse gases. Prominent ones are **carbon dioxide, methane, water vapor, nitrous oxide, and ozone**.

Greenhouse gases keep Earth from freezing over: without them, the average surface temperature would be about **60 °F colder**. About **2.4 billion years ago**, rising oxygen likely destroyed much of the warming methane, and Earth froze over almost completely. Too much

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greenhouse gas would melt the **ice over land** and expand the warming seawater, raising sea levels dangerously.

† The episode says greenhouse gases absorb the infrared radiation Earth "reflects." Earth emits (radiates) infrared because it is warm. Reflection is the albedo part of the story: sunlight bounced straight back to space.

Part 6 — Differential Heating, Global Winds, and Temperature

Because Earth is roughly spherical, sunlight strikes the equator much more directly than the poles, and because Earth's axis is tilted, the pattern shifts with the seasons. Earth is **differentially heated**. Warm air rises and cool air sinks, so this sets up **convection cells**, circular loops of air, with **low pressure** below rising warm air and **high pressure** below sinking cool air.

Each hemisphere has three cells from equator to pole: the **Hadley**, **Ferrel**, and **polar** cells. The jet streams form near their boundaries, where sharp temperature contrasts combine with the **Coriolis Effect**, an apparent force caused by Earth's rotation. In the Northern Hemisphere it deflects moving air to the air's right; in the Hadley cell, high-altitude northbound winds shift east and southbound surface winds shift west, becoming the **trade winds**. The effect is negligible at the equator and grows toward the poles.

Temperature is an indirect measure of the average kinetic energy of randomly moving molecules. It is not the same as heat; in the very thin thermosphere, a high temperature would not make you feel warm. **Fahrenheit** puts water's freezing and boiling points at **32° and 212°**, **Celsius** at **0° and 100°**, and **C = 5/9 (F − 32)**.

Part 7 — Volcanoes, Human Activity, and Climate

Volcanoes mostly affect climate in the short term. An eruption can send fine ash and **sulfur dioxide** into the stratosphere. The ash falls out within weeks, but the sulfur dioxide forms **sulfate aerosols** that linger for a year or two, reflecting sunlight and cooling Earth. **Mount Pinatubo** (Philippines, **1991**) lowered the next year's global temperature by about half a Celsius degree, roughly one Fahrenheit degree. Volcanoes also release carbon dioxide, **a few hundred million metric tons per year**, but that is only about **1%** of what humans add by burning fossil fuels.

Over the last **150 years** of data, human activity has greatly increased Earth's average temperature. Burning fossil fuels, deforestation, and raising methane-producing animals have caused a sharp rise in greenhouse gases and a sustained long-term warming. Sea levels are projected to rise roughly **1 to 3 feet (a third of a meter to 1 meter)** this century, and potentially several meters over the coming centuries.

Climate change doesn't just mean warmer temperatures. Adding thermal energy **supercharges** the thermodynamic processes that drive weather, bringing more intense storms, more severe droughts, more extreme precipitation, and less predictable weather. Because the ingredients are interconnected, a change in one cascades through the whole system.

From the first raindrop to the global winds, every part of Earth's weather is physics at work: heat on the move, driven by the Sun and shaped by water, rotation, volcanoes, and life itself.

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TRY IT YOURSELF

Count the storm, convert the forecast. You see a lightning flash and count 15 seconds before you hear the thunder. About how far away was the strike, in miles and in kilometers? Later, the forecast high is 86 °F. Use C = 5/9 (F − 32) to convert it to Celsius.

(Answer: at about 1/5 mile or 1/3 kilometer per second, 15 seconds means about 3 miles, or about 5 kilometers. For the forecast, C = 5/9 × (86 − 32) = 5/9 × 54 = 30 °C. A warm summer day, and a storm close enough to head indoors.)

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STUDY QUESTIONS

Questions and Model Answers

Model answers follow each question. Accept answers in students' own words that capture the key ideas.

Red numbers mark core questions — the ones to be sure students can answer.

Part 1 — What Weather Is

1. In which layer of the atmosphere does all Earth-based weather occur? What do meteorologists usually mean by "atmosphere" when they talk about weather?

> **Answer:** The troposphere. In the context of weather, "atmosphere" refers specifically to the troposphere.

2. From a physics standpoint, what is a thermodynamic event, and why can weather be described as a set of them?

> **Answer:** A thermodynamic event is driven by the movement of heat due to temperature differences. Weather is the result of energy transfer and energy transformation as heat moves through the atmosphere, so it fits this description.

3. List the six main ingredients that combine to create Earth-based weather.

> **Answer:** The atmosphere itself, water in all three phases, the Sun, Earth's rotation, volcanic activity, and biological activity.

4. Where does the word meteorology come from, and what does it literally mean?

> **Answer:** From the Greek *meteoros*, meaning "lofty" or "high in the air." Meteorology literally means "the study of things high in the air."

5. The Moon receives sunlight and has large temperature changes. Why doesn't it have weather?

> **Answer:** It lacks a substantial atmosphere, so there is no fluid to carry the motions (fluid dynamics) that weather systems need.

Part 2 — Water: Precipitation, Hail, Lightning, and Thunder

1. What makes water unique among the common substances in Earth's atmosphere?

> **Answer:** It is the only common substance that naturally exists in all three phases (ice, liquid water, and water vapor) within the temperature and pressure ranges of Earth's atmosphere.

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2. Describe the difference between rain, snow, and sleet.

> **Answer:** Rain is liquid water. Snow is fully formed ice crystal structures. Sleet is small pellets of ice that form when falling raindrops freeze before they reach the ground.

3. Describe how hailstones form. About how large can they get, and what is the U.S. record?

> **Answer:** In violent storms with cumulonimbus clouds, strong updrafts carry ice particles up and down through regions of supercooled water droplets, which freeze onto them layer by layer. Hailstones range from about 5 mm to more than 15 cm across; the U.S. record is 8 inches (about 20 cm).

4. Explain how electric charge builds up in a thunderstorm cloud and what lightning is.

> **Answer:** Collisions between ice crystals and graupel (soft hail pellets) separate electric charge, building enormous voltage differences between parts of the cloud and between the cloud and the ground. Lightning is the giant electrical discharge that dissipates that voltage difference, heating and ionizing the air along its path.

5. How does lightning help fertilize vegetation?

> **Answer:** Lightning's energy breaks up nitrogen molecules in the surrounding air, forming nitrogen compounds that dissolve in rainwater as nitrates, which fertilize the vegetation below.

6. What causes thunder? If you count 10 seconds between a lightning flash and its thunder, about how far away was the strike?

> **Answer:** Thunder is the sound of the shock wave made when lightning heats the air so suddenly that it expands explosively. At about 1/5 mile (1/3 km) per second, 10 seconds means about 2 miles, or about 3.3 km.

Part 3 — Clouds and Evaporation

1. What are condensation nuclei? Give three examples.

> **Answer:** Tiny airborne particles around which cloud droplets and ice crystals form. Examples: dust, smoke, sulfate particles, and bacteria.

2. What is the immediate reason cloud droplets form at a particular altitude?

> **Answer:** The relative humidity of the air there has reached 100%, because the air's temperature has dropped to the dew point for air at that altitude.

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3. Explain why evaporation is a cooling process.

> **Answer:** Only the fastest-moving molecules at the water's surface have enough kinetic energy to escape into the air. They carry that energy away, so the average kinetic energy (temperature) of the remaining liquid drops.

4. List at least four factors that affect how fast a puddle evaporates.

> **Answer:** Any four of: how large and shallow the puddle is; how warm the surrounding air is; how warm the ground beneath it is; whether air is blowing over it; whether it is in direct sunlight; how low the relative humidity of the surrounding air is.

5. What is latent heat, and how does its release help a cloud stay aloft and grow?

> **Answer:** It is the energy evaporating water molecules carried away from the liquid. When the vapor condenses into droplets or ice crystals, that energy is released, warming the air. The warmer, more buoyant air fuels updrafts that keep the cloud aloft and growing.

6. How many basic cloud types do meteorologists recognize, and what is fog?

> **Answer:** Ten basic types, called genera. Fog is simply a cloud at ground level.

Part 4 — Humidity and Dew Point

1. Define relative humidity and write the formula for it.

> **Answer:** At a given temperature, relative humidity is the ratio of the current water vapor pressure to the saturation (maximum) water vapor pressure, expressed as a percentage: RH = (current vapor pressure ÷ saturation vapor pressure) × 100%.

2. What does saturation vapor pressure depend on, and how?

> **Answer:** Almost entirely on temperature: the warmer the air, the higher the saturation vapor pressure.

3. What is the dew point? What happens to the relative humidity of a parcel of air if it warms up?

> **Answer:** The dew point is the lower temperature at which air, if cooled to it, would reach 100% relative humidity. If the air warms, its relative humidity decreases, because warmer air has a higher saturation vapor pressure.

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4. Why do we often find dew on grass or cars in the early morning?

> **Answer:** Overnight the air and surfaces cool. When grass, bushes, or a car cool to the dew point of the air around them, water vapor condenses onto them as dew.

5. Why is indoor air often very dry in winter, even when it is snowing outside?

> **Answer:** Cold outside air contains very little water vapor in absolute terms, even near 100% relative humidity. When it comes inside and warms, its saturation vapor pressure rises enormously, so its relative humidity drops sharply.

Part 5 — The Sun, Albedo, and Greenhouse Gases

1. What supplies nearly all the heating behind Earth's weather? Describe the two ways that energy reaches the atmosphere.

> **Answer:** The Sun, a nearly constant and renewable source of radiant energy. Some radiant energy warms the atmosphere directly; the rest warms Earth's lands and seas, which in turn warm the atmosphere.

2. What is albedo, and about what fraction of incoming sunlight does Earth reflect straight back into space?

> **Answer:** Albedo is Earth's reflectivity coefficient. About 30% of incoming sunlight is reflected straight back into space.

3. What is a greenhouse gas? Why are nitrogen and oxygen not greenhouse gases? Name three greenhouse gases.

> **Answer:** A gas in the atmosphere that absorbs and emits infrared radiation. Nearly all greenhouse gases have three or more atoms per molecule, and N_2 and O_2 have only two. Any three of: carbon dioxide, methane, water vapor, nitrous oxide, ozone.

4. Why are greenhouse gases necessary? What likely happened about 2.4 billion years ago?

> **Answer:** Without them Earth's average surface temperature would be about 60 °F colder, well below freezing. About 2.4 billion years ago, rising oxygen destroyed much of the methane keeping Earth warm, and Earth likely froze over almost completely.

5. Give two reasons too much greenhouse gas would raise sea levels.

> **Answer:** Melting ice that sits on land adds water to the oceans, and seawater expands as it warms (thermal expansion).

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Part 6 — Differential Heating, Global Winds, and Temperature

1. Why is Earth heated unevenly, and why does that pattern change through the year?

> **Answer:** Earth is roughly spherical, so sunlight strikes the equator much more directly than the poles. Earth's axis is tilted relative to its orbital plane, so the heating pattern shifts with the seasons.

2. How does differential heating create convection cells? Where do low- and high-pressure areas form?

> **Answer:** Warm air rises and cool air sinks, forming circular loops of air (convection cells). Areas below rising warm air become low-pressure areas; areas below sinking cool air become high-pressure areas.

3. Name the three convection cells in each hemisphere, in order from the equator. Where do the jet streams form?

> **Answer:** The Hadley cell, the Ferrel cell, and the polar cell. The jet streams form near the boundaries between the cells, where sharp temperature contrasts combine with the Coriolis Effect.

4. What is the Coriolis Effect? Which way does it deflect moving air in the Northern Hemisphere, and how does it produce the trade winds?

> **Answer:** An apparent ("fictitious") force caused by Earth's rotation. In the Northern Hemisphere it deflects moving air to the air's right. In the Hadley cell, northbound high-altitude winds are shifted east and southbound surface winds are shifted west, becoming the trade winds. The effect is negligible at the equator and grows stronger toward the poles.

5. Define temperature. Why wouldn't the very high temperatures of the thermosphere make you feel warm?

> **Answer:** Temperature is an indirect measure of the average kinetic energy of a substance's randomly moving molecules. It is not the same as heat or internal energy; the thermosphere's air is so thin that there are too few molecules to transfer much energy to you. How warm you feel depends on the density of the air as well as its temperature.

6. Use $C = 5/9\ (F - 32)$ to convert 68 °F to Celsius. Show your work.

> **Answer:** $C = 5/9 \times (68 - 32) = 5/9 \times 36 = 20$ °C.

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Part 7 — Volcanoes, Human Activity, and Climate

1. How can a large volcanic eruption cool Earth in the short term? Why doesn't the ash cause most of the cooling?

> **Answer:** The eruption sends sulfur dioxide into the stratosphere, where it forms sulfate aerosols that linger for a year or two and reflect incoming sunlight. The ash falls out within weeks, so it doesn't last long enough to matter much.

2. What effect did the 1991 eruption of Mount Pinatubo have on global temperature?

> **Answer:** It reduced the average global temperature the following year by about half a Celsius degree, roughly one Fahrenheit degree.

3. How does the carbon dioxide released by all volcanic activity compare with the amount released by humans?

> **Answer:** Volcanoes add a few hundred million metric tons of CO_2 per year, only about 1% of what humans add each year by burning fossil fuels.

4. List three human activities that increase greenhouse gases. How much is sea level projected to rise this century?

> **Answer:** Any three of: burning fossil fuels, deforestation, farming methane-producing animals for human consumption (and other human activities). Sea level is projected to rise roughly 1 to 3 feet (about 0.3 to 1 m) this century.

5. Why does climate change mean more extreme weather, not just warmer temperatures?

> **Answer:** Greenhouse gases add thermal energy to the atmosphere, supercharging the thermodynamic processes that drive weather. The result is more intense storms, more severe droughts, more extreme precipitation, and less predictable weather.

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FOR THE INSTRUCTOR

Corrections and Cautions

1. Earth emits infrared; it doesn't reflect it (about 10:20).

The episode says greenhouse gases absorb the infrared Earth "reflects." Earth radiates (emits) infrared because it is warm. Reflection is the albedo part: about 30% of sunlight bounced straight back to space. Students often merge the two ideas, so keep them separate. This is also flagged as a footnote under the Part 5 summary in both guides.

2. "Subsequently" should be "Consequently" (about 10:35).

"Subsequently, the two-atom nitrogen and oxygen molecules… are not greenhouse gases." The link is logical, not a sequence in time.

3. Where latent heat is released (about 07:33).

The episode says condensation warms the air "directly below" the cloud's droplets. The heat is released where condensation happens, warming the air inside the cloud and the rising air feeding it. Footnoted under the Part 3 summary in both guides.

4. The 2.4-billion-year freeze is a leading hypothesis.

Linking the Huronian glaciation to methane loss during the Great Oxidation Event is widely supported but not settled. The episode's word "likely" is appropriate; don't let students turn it into a certainty.

5. Greener grass after storms is only partly nitrates.

Lightning does fix nitrogen into nitrates, but the amount per storm is modest. Rain washing dust off leaves and hydrating plants also contributes to the effect.

6. "Nearly monotonic" describes the long-term trend.

Year-to-year global temperatures still rise and fall (El Niño, La Niña, eruptions like Pinatubo). Students may point to one cool year; the claim is about the multi-decade trend.

7. Only land ice raises sea level when it melts.

The episode correctly says "ice cover over land." Floating sea ice already displaces its own weight of water, so melting it barely changes sea level. This is a common misconception worth naming.

8. Coriolis deflection reverses in the Southern Hemisphere.

The episode gives only the Northern Hemisphere rule (deflection to the right). In the Southern Hemisphere moving air is deflected to its left.

9. "Centigrade" is the older name.

The scale was officially renamed Celsius in 1948. The episode's "also called Centigrade" is fine as history.

FOR THE INSTRUCTOR

Teaching Notes

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Connect to Episode 1.

Students have already met the troposphere, thermosphere, homosphere, and jet streams in 5AE1. Open with a two-minute recall of the layers before introducing the six ingredients.

Demo evaporative cooling.

A few drops of rubbing alcohol or water on the back of the hand, with a fan, make "evaporation is a cooling process" immediate. Link it back to latent heat being released later in the cloud.

Relative versus absolute humidity.

The winter-indoor-air example is the clearest way to show that 100% relative humidity in cold air can mean very little water. The "elbow room" analogy from the episode works well on the board.

Practice the conversion both ways.

Have students derive F = 9/5 C + 32 from the episode's formula, then check with the fixed points 0/32, 100/212, and the crossover at −40°.

Coriolis misconceptions.

Students often have heard that sinks and toilets drain in opposite directions across the equator. The effect is far too weak at that scale; it matters for large, long-lived air motions.

Temperature is not heat.

Use the thermosphere example: very high temperature, almost no molecules. Contrast reaching into a hot oven (air at 200 °C) with touching the oven rack.

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